

ATPL EASA STUDY GUIDE | 061 - GENERAL NAVIGATION

Chapter 3: Earth Magnetism, Deviation & The Direct-Reading Compass

Syllabus Source: CAE Oxford (pp. 85-128) | Pass Target: >= 90% AviationExam

1. EARTH'S MAGNETIC FIELD & TERMINOLOGY

The Earth behaves as a giant magnetic dipole whose magnetic poles are offset from the true geographic poles:

- **Magnetic Variation (Declination):** The angular difference between True North (TN) and Magnetic North (MN). Lines connecting points of equal variation are ISOGONALS. The line of zero variation is the AGONIC LINE.
- **Compass Deviation:** The angular error between Magnetic North and Compass North caused by internal aircraft magnetism (radios, engines, steel structure). Lines of equal deviation: Isoclinals.
- **The Navigation Master Chain:** Compass Heading (CH) +/- Deviation = Magnetic Heading (MH) +/- Variation = True Heading (TH) ('Cadbury Dairy Milk Very Tasty': C -> D -> M -> V -> T).
Variation/Deviation West is ADDED; East is SUBTRACTED when converting Compass to True!

2. DIRECT-READING COMPASS DYNAMIC TURNING & ACCELERATION ERRORS

Due to magnetic dip (the magnetic needle wants to tilt towards the ground), the center of gravity of the magnet is weighted, creating dynamic errors in the Northern Hemisphere:

- **Turning Errors (UNOS Rule):** 'Undershoot North, Overshoot South'. When turning THROUGH NORTH, the compass leads and turns ahead; you must ROLL OUT EARLY (undershoot). When turning THROUGH SOUTH, the compass lags; you must ROLL OUT LATE (overshoot)!
- **Acceleration Errors on E/W Headings (ANDS Rule):** 'Accelerate North, Decelerate South'. When accelerating on an East or West heading, the compass swings NORTH; when decelerating, the compass swings SOUTH!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Compass Acceleration & Turning

Acceleration Errors (ANDS Rule - East/West Headings in Northern Hemisphere): -
 Accelerate -> Pendulous mass lags, compass falsely indicates a turn toward NORTH. -
 Decelerate -> Compass falsely indicates a turn toward SOUTH. Turning Errors (UNOS Rule - Turning onto Northerly or Southerly Headings): - Turning onto NORTH: Compass leads the turn -> Undershoot North (stop roll-out early). - Turning onto SOUTH: Compass lags behind -> Overshoot South (roll out late past the desired heading).

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)

[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Complete Compass to True Heading Conversion

SCENARIO (Fundamental Flight Deck Navigation Drill): A pilot reads the direct-reading standby magnetic compass: - Compass Heading (CH) = 085° - Compass Deviation Card indicates: For 090°, Deviation is -3° (3° West) - En-route chart shows local Magnetic Variation = 7° East - Wind drift correction applied: Drift is 5° to the Right

QUESTION: Calculate (1) Magnetic Heading, (2) True Heading, and (3) True Track.

SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Convert Compass Heading to Magnetic Heading (Apply Deviation): - Rule: Deviation West is SUBTRACTED when going Compass -> Magnetic? CAUTION! - Memory Verse: 'Deviation West, Magnetic Best' -> Magnetic is BIGGER (ADD West Deviation)! - CH (085°) + Dev West (3°) = MH 088°

Magnetic! Step 2: Convert Magnetic Heading to True Heading (Apply Variation): - Variation is 7° East. - Memory Verse: 'Variation East, Magnetic Least' -> Magnetic is SMALLER (True is BIGGER, so ADD East Variation)! - True Heading = MH (088°) + Var East (7°) = 095° True!

Step 3: Calculate True Track (Apply Wind Drift): - True Heading = 095° - Wind pushes aircraft 5° to the RIGHT (+5° Drift) - True Track = True Heading + Drift = 095° + 5° = 100° True!

FINAL ANSWER: Magnetic Heading = 088°; True Heading = 095°; True Track = 100°.

[!] EXAM TRAP (AVIATIONEXAM) - The UNOS Turning Rule Trap

- Turning through NORTH in Northern Hemisphere: Roll out 20° to 30° BEFORE the compass heading (UNDERSHOOT)! - Turning through SOUTH: Roll out 20° to 30° AFTER the compass heading (OVERSHOOT)! - In the Southern Hemisphere, the rules are inverted (USON: Undershoot South, Overshoot North)!

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In the Northern Hemisphere, an aircraft is flying on a heading of 090° (East) and accelerates. What will the

- [A]: A turn towards South
- [B]: A TURN TOWARDS NORTH (ANDS: Accelerate North, Decelerate South)
- [C]: It remains stationary
- [D]: It oscillates 180°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Acceleration forces the pendulous compass weight backwards, tilting the needle so the vertical component of Earth's magnetic field pulls the north needle towards the north.

Q2: What is the name of a line on an aeronautical chart connecting points of ZERO MAGNETIC VARIATION?

- [A]: Isogonal
- [B]: AGONIC LINE
- [C]: Isocline
- [D]: Aclinic line

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. An isogonal is a line of equal variation; the specific isogonal of 0° variation is called the agonic line.

Q3: If Compass Heading is 320°, Deviation is 2°W, and Variation is 12°W, what is the TRUE HEADING?

- [A]: 334°
- [B]: 306° (West is best: MH = 320 - 2° No, CH + Dev = MH. 320 - 2 = 318 - 12 = 306° True)
- [C]: 320°
- [D]: 340°

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. From Compass to True: CH 320° - 2°W Dev = 318° MH - 12°W Var = 306° True.